

SOLUTION

The Eight Puzzle Problem is solved using a **state-space search** approach, where each arrangement of tiles represents a state and each valid tile movement creates a new state. The solution process is as follows:

1. Represent the puzzle as a 3×3 **matrix** with one blank space (0).
2. Define the **initial state** and the **goal state**.
3. Identify all possible moves of the blank tile (Up, Down, Left, and Right).
4. Generate new states by moving the blank tile to a valid adjacent position.
5. Compare each generated state with the goal state.
6. Avoid revisiting previously explored states to prevent repeated processing.
7. Continue exploring new states until the goal state is reached or no further states are available.
8. Once the goal state is found, display the sequence of moves leading to the solution.